

## Problem Set 1

Due: Monday, Sept 14, at 9 am. Late problem sets are not accepted.

Problem set solutions should show all necessary steps and be clearly explained and presented. Students with poor handwriting may need to deliberately rewrite or typeset their solutions using programs such as LaTeX or LyX. I anticipate and encourage that you will work together on problem sets. Productive collaboration is an important part of science. However, the work you turn in should be your own, and all collaborators should be acknowledged. Copying work will result in no credit for all involved. While you are encouraged to use whatever static sources you want (i.e. textbooks, class notes, scholarly papers, etc. whether in print or online), responsive sources such as posts on online fora or using chatGPT or other generative AI is disallowed unless explicitly allowed by me. Homework is not assigned simply to accumulate grades, but is the way you learn through practice - using AI to find solutions is no different than copying answers from others - both undermine the learning process. If you need an accommodation for a religious observance, due to sickness or a personal or family emergency, or a disability, contact me as early as possible, and I will work with you to find a solution.

With each problem, I'm providing a rough degree-of-difficulty estimate.

(1) indicates a “warm up” problem; (2) indicates a standard problem: straightforward but perhaps taxing; (3) a challenging problem; (4) a “stumper” (open-ended) problem where you'll have to think about what approximations are reasonable and where you'll likely have to draw on other sources than the textbook. Problem (0) is my pet project. These are facts that should be imprinted indelibly on a physicist's brain.

### 0. (0) Series expansions

What are the polynomial series in small  $x$  for these functions:

$$(a) e^x \quad (b) \sin(x) \quad (c) \cos(x) \quad (d) \log(1+x) \quad (e) (1+x)^p \quad \text{and} \quad (f) \frac{1}{1-x}$$

to the third non-zero term? Know these six expansions<sup>1</sup> cold. **You will be expected to recall these from memory on exams.**

### 1. (1) Width of the binomial distribution

Show that for the binomial distribution

$$\langle n^2 \rangle = \left( p \frac{\partial}{\partial p} \right) \left( p \frac{\partial}{\partial p} \right) (p+q)^N = Np + N(N-1)p^2$$

and that the standard deviation is  $\sqrt{Npq}$ .

### 2. (1) Poisson processes

a) Show that  $P(n; t) = \frac{(rt)^n}{n!} \exp(-rt)$  is the solution to

$$\frac{dP(n; t)}{dt} = r [P(n-1; t) - P(n; t)]$$

and is thus the probability of  $n$  events occurring in time  $t$  in a Poisson process with rate  $r$ .

b) Verify that  $1 = \sum_{n=0}^{\infty} P(n; t)$ .

c) Use  $P(n; t)$  to calculate the mean number of events,  $\langle n \rangle$ , and its standard deviation.

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<sup>1</sup>Recall that these are just the Taylor expansions which can be derived from

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

where primes indicate derivatives with respect to  $x$

### 3. (2) Comparison of distributions

NB: For this problem, I recommend using python (free to download and install) or mathematica (free for all UMD students). Since many of you may not be familiar with mathematica, I've provided a notebook that should help you get started defining functions and plotting them.

Often times, we are confronted with mathematical functions/distributions that look complicated. One way to get a handle on these functions is to simply plot them for a range of values. Sometimes, you can 'discover' properties of these distributions that aren't so obvious from their functional form.

a) Make the following two plots, with axes clearly labeled and functions identified by a legend (the labeling and legend can be added by hand if you don't want to program that in):

- i. On the same axes, plot the Gaussian with mean  $\mu_G = 1$  and standard deviation  $\sigma = 0.5$ ; the Binomial distribution with  $N_{samples} = 10$  and probability  $p = 0.1$ ; and the Poisson distribution with mean  $\mu_P = 1$ .

Hint: The independent variable (horizontal axis) is the value of a sample from the distribution, while the vertical axis is the probability of getting that value. For the binomial distribution, there's a restriction on the domain of the independent variable. Think about this and set the plot range accordingly.

- ii. On the same axes, plot the Gaussian with mean  $\mu_G = 2$  and standard deviation  $\sigma = 1$ ; the Binomial distribution with  $N_{samples} = 10$  and probability  $p = 0.2$ ; and the Poisson distribution with mean  $\mu_P = 2$ .

b) Discuss the plots from the previous part. Can you make a conjecture about a property of these distributions?

c) There's another interesting property to explore as we let the mean of all of these distributions increase. Make the following four plots, with axes clearly labeled and functions identified by a legend (the labeling and legend can be added by hand if you don't want to program that in):

- i. On the same axes, plot the Gaussian with mean  $\mu_G = 2$  and standard deviation  $\sigma = \sqrt{\frac{2}{2}} = 1$ ; and the Binomial distribution with  $N_{samples} = 4$  and probability  $p = 0.5$ .

Hint: The independent variable (horizontal axis) is the value of a sample from the distribution, while the vertical axis is the probability of getting that value. For the binomial distribution, there's a restriction on the domain of the independent variable. Think about this and set the plot range accordingly.

- ii. On the same axes, plot the Gaussian with mean  $\mu_G = 2$  and standard deviation  $\sigma = \sqrt{2}$ ; and the Poisson distribution with mean  $\mu_P = 2$ .
- iii. On the same axes, plot the Gaussian with mean  $\mu_G = 18$  and standard deviation  $\sigma = \sqrt{\frac{18}{2}} = 3$ ; and the Binomial distribution with  $N_{samples} = 36$  and probability  $p = 0.5$ .
- iv. On the same axes, plot the Gaussian with mean  $\mu_G = 18$  and standard deviation  $\sigma = \sqrt{18}$ ; and the Poisson distribution with mean  $\mu_P = 18$ .

d) Discuss the plots from the previous part. Can you make a conjecture about a property of these distributions?

#### 4. (2) Shot noise

Often, one tries to measure the intensity of light from a weak source using a photomultiplier tube. Incoming photons eject photoelectrons from a cathode. These are accelerated towards a “dynode” which emits several electrons for every electron which strikes it. By cascading these after each other, a single initial photoelectron can produce a single current pulse of  $\sim 10^6$  electrons which is temporally well resolved ( $\sim 5$  ns). For a fixed light source, the pulses arrive *at random* with an average rate  $r_s$  counts/second.

$$r_s = \eta \frac{IA}{h\nu} \quad (1)$$

where  $I$  is the intensity of the light (power per unit area),  $A$  is the photocathode area,  $\nu$  is the frequency of the light,  $h$  is Planck’s constant, and  $\eta$  is the quantum efficiency of the cathode (often  $\sim 20\%$ ).

- Find the mean number of counts expected in a time interval  $T$ . This is the signal  $S$ .
- If measurement is repeated for a number of time intervals each of the same length, different numbers of counts will be recorded. What is the standard deviation  $\sigma$  of the number of counts? This is referred to as the *shot noise*  $N$ .
- Find the signal to noise ratio  $S/N$ . How does it depend on the intensity  $I$  and the time  $T$ ?
- Thermal emission of electrons from the cathode causes pulses indistinguishable from the photopulses. They occur at a rate  $r_d$  and are referred to as *dark current* since they are independent of  $I$ . The number of counts can be recorded in a time  $T$  when  $I$  is on, and subtract from that the number counted in a time  $T$  when  $I$  is off. This difference is the signal  $S$  which is proportional to  $I$ . Find the signal to noise ratio for this experiment in the case where  $r_d \gg r_s$ . Now how does  $S/N$  depend on  $I$  and  $T$ ?

#### 5. (3) S1.39 Speed of sound

#### 6. (2+) S1.40 Atmospheric temperature gradient

#### 7. (3) Microbiology

A single bacterium is in some fluid doing those things that bacteria do. The bacterium needs oxygen (among other things) to survive. The oxygen is all around it, dissolved in water, with a concentration  $n_0$  of about  $0.2$  mol/m<sup>3</sup>. The oxygen near the bacteria is depleted by the bacteria as it uses it up. The oxygen concentration  $n(r)$  in the fluid depends on the distance to the center of the bacterium  $r$ . Very far away  $n(\infty) = n_0$  and we’ll assume that at the surface of the bacterium (radius  $R$ ) every oxygen is consumed immediately, thus  $n(R) = 0$ . From Fick’s law, there must be a flux  $J$  of oxygen inward.

- Imagine drawing a series of concentric shells around the bacterium with radii  $r_1$ ,  $r_2$ , etc. Oxygen crosses each shell on its way to the center. In a steady state, convince yourself that the number of oxygens crossing each shell is equal. Therefore the current  $I$  must be a constant equal to the flux  $J(r)$  times the surface area of the shell. Write an expression for  $J(r)$  in terms of the constant  $I$ .
- Write Fick’s law and solve it for  $n(r)$ . Remember to use boundary conditions at  $r = R$  and  $\infty$ .
- What is  $I$ ? Given  $D \approx 10^{-9}$  m<sup>2</sup>/s and  $R = 1$   $\mu$ m, get a numerical answer. You have just computed, without using any biochemistry at all, the maximum rate at which oxygen can be consumed by any bacteria.
- A measure of an organism’s overall metabolic activity is its rate of  $O_2$  consumption divided by its mass. Find the maximum possible activity of a bacterium of radius  $R$ .
- The actual oxygen consumption is  $0.02$  mol/kg s. What limit do you get on the size  $R$  of a bacterium?
- Can you think of a way for a bacterium to evade the limit in (e)?

## 8. (4) A winter night

As the sun sets over a pond, the water is uniformly at  $0^\circ\text{C}$  and no ice is present. When night falls, the air temperature immediately drops to some temperature,  $T$ , below freezing. Our goal is to develop a model that predicts the thickness of ice on the pond as a function of the average surface air temperature above the pond  $T < 0^\circ\text{C}$ . [Heat is being transferred to the air at temperature  $T$  through the ice layer of thickness  $x(t)$  from still water below at  $0^\circ\text{C}$ .] Relate the heat lost to the water-to-ice transformation and the cooling of ice. Solve the differential equation for  $x$ . How does your result depend on  $T$ ? With values for the thermal conductivity  $k_t$ , the latent heat of formation  $L$ , and specific heat  $c$  of ice, give a quantitative prediction for how long it will take to get enough ice to skate over ( $\approx 2$  cm) if  $T = -5^\circ\text{C}$ . Any comments on our assumptions?